DSA REFERENCE

Union-Find

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Maintain disjoint components under repeated merges with parent links, path compression, and union by rank or size.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / UNION-FIND

DSA

Union-Find

CORE IDEA

Maintain disjoint components under repeated merges with parent links, path compression, and union by rank or size.

Recognition cues

+ dynamic connectivity
+ redundant edge
+ component count after unions
+ Kruskal minimum spanning tree

Implementation sequence

+ initialize each element as its own root
+ compress during find
+ merge roots not raw nodes
+ update count only on a real merge

Invariant

Every element reaches exactly one representative; component metadata belongs only to representatives.

Worked reasoning

An edge whose endpoints already share a representative is redundant and closes a cycle.

Complexity

Nearly O(1) amortized per operation: O(alpha(n)); O(n) storage.

Failure modes

- [ ] merging non-root nodes
- [ ] incorrect rank updates
- [ ] counting duplicate unions
- [ ] using it when deletions or path details are required

Practice ladder

Union-Find: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Number of Provinces**	State the cue, invariant, edge case, and proof before implementation.
2	**Redundant Connection**	State the cue, invariant, edge case, and proof before implementation.
3	**Accounts Merge**	State the cue, invariant, edge case, and proof before implementation.
4	**Graph Valid Tree**	State the cue, invariant, edge case, and proof before implementation.
5	**Most Stones Removed**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.